

SML2026

Practice

Q1. Consider two Cauchy distributions in one dimension, where the probability density function (PDF) for each class is given by:

$$p(x | \omega_i) = \frac{1}{\pi b \left(1 + \left(\frac{x-a_i}{b}\right)^2\right)}, \quad i = 1, 2,$$

where:

- a_1 and a_2 are the location parameters ($a_2 > a_1$),
- b is the scale parameter (common for both distributions),
- $P(\omega_1) = P(\omega_2) = 0.5$ (equal priors).

The Neyman-Pearson criterion is applied with a zero-one loss function, where the maximum acceptable error rate for misclassifying a sample from ω_1 as belonging to ω_2 is E_1 .

Tasks

1. **Decision Boundary:** Determine the decision boundary in terms of the parameters a_1 , a_2 , b , and E_1 .
2. **Error Rate for ω_2 :** For the derived decision boundary, compute the error rate for misclassifying a sample from ω_2 as belonging to ω_1 , denoted as $P(\text{error} | \omega_2)$.
3. **Overall Error Rate:** Compute the overall error rate $P(\text{error})$ under the zero-one loss assumption:

$$P(\text{error}) = 0.5 \cdot P(\text{error} | \omega_1) + 0.5 \cdot P(\text{error} | \omega_2).$$

4. **Specific Case:** Apply the results to the specific case where:

- $b = 1$,
- $a_1 = -1$,

- $a_2 = 1$,
- $E_1 = 0.1$.

Compute:

- The decision boundary.
- The error rate $P(\text{error} \mid \omega_2)$.
- The overall error rate $P(\text{error})$.

Q2.

Consider a binary classification problem where a random variable x follows one of two Cauchy distributions

$$p(x \mid \omega_i) = \frac{1}{\pi b \left(1 + \left(\frac{x-a_i}{b}\right)^2\right)}, \quad i = 1, 2,$$

and:

- a_1 and a_2 are the location parameters ($a_2 > a_1$),
- b is the common scale parameter for both distributions,
- $P(\omega_1) = P_1$ and $P(\omega_2) = P_2$ are the prior probabilities, with $P_1 + P_2 = 1$.

Tasks

- Find the discriminant function for each class.
- Find the decision boundary.
- Specific Case:** Apply the results to the following parameters:

- $a_1 = -1$,
- $a_2 = 1$,
- $b = 1$,
- $P_1 = 0.7$, $P_2 = 0.3$.

- Compute the decision boundary x^* . An analytical form may not be feasible.
- Determine the class decision for $x = 0$ and $x = 2$.

Q3.

The following are the 2×3 matrices for the three classes:

Data matrix for Class 1 (A)

$$A = \begin{bmatrix} 0.12 & 0.45 & 0.78 \\ 0.34 & 0.56 & 0.91 \end{bmatrix}$$

Data matrix for Class 2 (B)

$$B = \begin{bmatrix} 0.23 & 0.67 & 0.89 \\ 0.45 & 0.12 & 0.34 \end{bmatrix}$$

Data matrix for Class 3 (C)

$$C = \begin{bmatrix} 0.98 & 0.76 & 0.11 \\ 0.33 & 0.44 & 0.55 \end{bmatrix}$$

Assume dimension to be 2. Let the data be distributed according to MVG.

- Compute likelihood for each class.
- Compute the prior.
- Using discriminant analysis, find to which class does a point $x = [0, 0]^\top$ belong to.
- Find the decision boundary between classes A and B.

Problem Statement

Consider a classification problem with 3 classes ($\omega_1, \omega_2, \omega_3$) and 4 independent binary-valued features (x_1, x_2, x_3, x_4):

- For each class ω_j ($j = 1, 2, 3$):

$$P(x_i = 1 \mid \omega_j) = p_{ji}, \quad P(x_i = 0 \mid \omega_j) = 1 - p_{ji}, \quad \text{for } i = 1, 2, 3, 4.$$

- The class priors are:

$$P(\omega_1) = 0.4, \quad P(\omega_2) = 0.35, \quad P(\omega_3) = 0.25.$$

- The feature probabilities are:

$$\text{For } \omega_1 : \quad p_{11} = 0.8, p_{12} = 0.6, p_{13} = 0.7, p_{14} = 0.5,$$

$$\text{For } \omega_2 : \quad p_{21} = 0.4, p_{22} = 0.8, p_{23} = 0.6, p_{24} = 0.7,$$

$$\text{For } \omega_3 : \quad p_{31} = 0.5, p_{32} = 0.4, p_{33} = 0.9, p_{34} = 0.3.$$

- Suppose you observe the feature vector:

$$X = (x_1, x_2, x_3, x_4) = (1, 0, 1, 1),$$

where $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 1$.

Discriminant Function Formulation

Define the discriminant function for each class ω_j as:

$$g_j(X) = \ln P(\omega_j) + \sum_{i=1}^4 [x_i \ln p_{ji} + (1 - x_i) \ln(1 - p_{ji})].$$

The classification decision is made by selecting the class ω^* that maximizes the discriminant function:

$$\omega^* = \arg \max_j g_j(X).$$

Tasks

1. **Compute the Discriminant Functions:** For each class ω_j , compute $g_j(X)$ using the given probabilities for $P(\omega_j)$ and p_{ji} .
2. **Compare the Discriminants:** Compare $g_1(X), g_2(X), g_3(X)$ and determine the class X belongs to.
3. **Numerical Example:**

- For the given feature vector $X = (1, 0, 1, 1)$, compute:

$$g_1(X), g_2(X), g_3(X).$$

- Determine the class ω^* to which X belongs.

Expected Results

- Compute the discriminant values $g_1(X), g_2(X), g_3(X)$.
- Identify the class ω^* that maximizes the discriminant function.